

Nuju Workshop — Summer Meju, Week 1

Pea Meju

Date: June 29, 2026, 2:00–5:00 PM

Venue: Mak Seoul

Instructor: Lee Kyung-ja (CEO, Hongju Foods, Hongseong)

Part 1: Summary

This lecture is the first session of a workshop on Korean traditional fermentation culture, focusing in particular on pea meju and the meju/jang made from a variety of ingredients. Instructor Lee Kyung-ja introduced the historical background of meju, jang-making using diverse legumes and alternative ingredients, summer meju production methods (pea meju, cheongtaejang, etc.), and lesser-known traditional jang such as sangsiljang (acorn meju jang) and taegakjang (soybean-husk soy sauce). The purpose of this workshop is to revive Korean food culture, which was severed during the Japanese colonial period and the turbulent times that followed, and to promote R&D of Korean fermentation culture through collaboration with American researcher Jason White. The program consists of four weeks total, and this session included hands-on time for participants to make pea meju themselves.

1. Workshop Background and Purpose

(1) The Rupture and Recovery of Korean Traditional Food Culture

1. Although meju is commonly perceived as something made in December/winter, records of making meju in summer exist from the 1800s (500 years ago), and summer meju was common knowledge in the past.
2. During the 40 years of Japanese colonial rule (1905–1945), Korean traditional culture was suppressed and severed; and in the roughly 40 years afterward (1945–1985), there was no time to spare for studying tradition. This roughly 80-year rupture meant our food culture could not be carried forward.
3. Recently, as economic strength and cultural confidence have been restored, a movement to rediscover the wisdom of our ancestors and traditional food culture has emerged.

(2) The Specific Purpose of the Workshop and the Collaboration Background

4. Meju and doenjang are the most central ingredients of Korean fermentation; re-examining them and finding better methods is the aim of the workshop.
5. Jason White, from the United States, has begun R&D on various fermented foods including Korean fermentation, and is looking for colleagues and supporters to share the research process with Koreans and build an R&D lab together.
6. This workshop was held to aid the research by sharing the Korean perspective and approach with Jason.

(3) The Four-Week Curriculum

- Week 1 (current session, as of 2026-06-29): Pea meju and summer meju
- Week 2: Barley bran meju
- Week 3: Integrated review centered on the content of old cookbooks (gojoriseo)
- Week 4: The JDC team (Jason White's side) joins the whole process, presenting and discussing their impressions and findings

Other traditional fermented foods such as nuruk (fermentation starter) and vinegar will be covered sequentially in the future.

2. Introduction to Various Jang and the Production Philosophy

(1) The Background to the Variety of Jang (the Instructor's Grandmother's Story)

7. The instructor's maternal grandmother was forcibly relocated to Manchuria during the Japanese colonial period. The Japanese ransacked all the grain storehouses and took the food, but the grandmother hid various seeds—cabbage, lettuce, green-onion seeds, and more—inside a small pillow.
8. Because the Japanese did not find these seeds in the pillow, the seeds were preserved. After the forced relocation, she sowed these seeds in the barren land of Manchuria, and the fertile soil yielded abundant harvests, with cabbages growing large.
9. This experience became the root of creative attempts to make jang from a wide variety of ingredients.

(2) Various Types of Beans and Jang Ingredients

10. Korea has about 300 native legume varieties, including the kidney-bean lineage, the soybean lineage, the red-bean lineage, and more.

11. Generally, jang beans (soybeans) are most commonly used for jang-making, but when heavy rain prevents soybeans from growing, alternative ingredients are needed.
12. Meju and jang can also be made from various legumes such as red beans, peas, mung beans, lentils, kidney beans, and sword beans.
13. Ingredients not found in Korea, such as lentils, can be sourced locally by overseas chefs to make their own jang.

(3) Why Quick Jang (Geupjang) Exists

14. Quick jang (geupjang), which can be made and eaten rapidly, developed when ingredients were scarce due to climate anomalies, invasions, poverty, etc.
15. Because jang was needed even in summer, summer quick jang such as pea meju came into being.

3. Detailed Introduction to Special Jang

(1) Sangsiljang (Acorn Meju Jang)

16. A jang using meju made from acorns; it was originally developed by monks in environments without soybeans (mountainous regions of North Korea, temples, etc.).
17. Acorns are ground, soaked in water to leach out the astringency (tannins), then mixed with soybeans, made into meju, and submerged in jang.
18. The flavor is very distinctive and excellent; it is said there were many cases of sick people coming to temples, eating this food, and recovering before returning home.
19. Ganggye (Ganggyejang), corresponding to the “frontal lobe” (northern region) of the Korean peninsula map, is famous for sangsiljang.
20. Acorns contain antibacterial components, which made it difficult for mold to develop well during meju-making, but after many attempts, proper meju production was achieved.
21. It won the Grand Prize at the 2022 Cham Fermentation Awards.
22. According to the literature, acorns improve immunity and have detoxifying functions, and are also known to be effective in relieving post-surgery cognitive impairment caused by anesthesia (memory decline, forgetting passwords, etc.). For this reason, there is a custom of giving or eating acorn jelly or sangsiljang doenjang porridge after

surgery.

(2) Taegakjang (Soybean-Husk Soy Sauce)

23. A soy sauce made by fermenting soybean husks (taegak) in brine.
24. Historical background: When wealthy households shelled soybeans to cook and discarded the husks, the poor people working in those houses gathered the unripe beans and husks and soaked them in brine to make jang to eat.
25. Chewing soybean husks reveals sweetness and depth, and fermenting them in brine yields a richly flavored soy sauce.
26. In production, the husks are washed well and sterilized by boiling above 100°C for over 30 minutes, then fermented.
27. As fermentation proceeds, impurities float to the top; skimming these off cleans it up neatly.
28. After completion, it is heated to about 70°C to kill only the microbes and stop fermentation (it must not be boiled above 100°C).

(3) Cheongtaejang (Green-Soybean Summer Jang)

29. Green soybeans are steamed, pounded in a mortar to make meju, pressed into cups, and fermented in sunlight or a hot place.
30. If submerged in jang without much salt, the taste is light and good, but since it is summer and insects appear easily, it is important to dry the meju thoroughly.
31. Because fermentation proceeds quickly, it is made small and processed swiftly.
32. If the flavor is shallow due to low salt, mixing in a little aged doenjang or good soy sauce brings out depth.
33. The current method was composed by referring to the cheongtaejang recipe of Master Bingheogak (a female practical scholar of the late Joseon period), recorded in the old cookbook Gyuhap Chongseo.

4. Pea Meju Production and Hands-On Practice

(1) Characteristics of Pea Meju and Why It Is Made Small

34. A summer meju made from peas; the reason it is made small (walnut-sized) is that insects like meju in summer.
35. The meju is wrapped in wrapping leaves (chojae) that have an aroma insects dislike, to prevent insects.

(2) Types and Roles of Wrapping Leaves (Chojae)

36. Cocklebur leaf: Used to wrap the meju like a grape. It has an aroma insects strongly dislike and retains moisture, helping the meju ferment well.
37. Lotus leaf: Aromatic, it can be used as a wrapping material that envelops the meju neatly like pottery.
38. Pine needles: Placed between meju here and there to repel insects.
39. Bamboo inner stalk: Removing the bamboo leaves and using only the inner stalk to wrap the meju is good for fermentation. In regions with abundant bamboo, making nuruk inside bamboo yields tasty jang.
40. Reeds and wheat stalks: When cut and dried, useful microbes such as yellow koji mold inhabit them and can be used for fermentation.

(3) Pea Meju Fermentation Method

41. Wrap in chojae and first dry the surface for one day (surface-drying), then wrap completely in leaves.
42. Place the wrapped meju in a sunny, warm spot; the warm heat of midsummer makes the meju ferment well.
43. Ferment for about one week.

(4) Hands-On Process

44. Participants shell the peas themselves, put them in a mortar, and pound them.
45. When using the mortar, fill only about halfway and work in pairs.
46. Reason for removing the husks: with husks, air pockets form and the degree of compression varies, making the meju sizes uneven.
47. Each participant makes six walnut-sized meju from the pounded beans to take home.
48. A demonstration was also held in which 250 g of pea meju the instructor had made in advance was submerged with water and aged doenjang for a taste test.

5. Making Brine

(1) Critical-Point (Saturated Brine) Method

49. Rather than calculating the ratio of salt to water numerically, on a large-volume basis (2–3 tons), salt (4–5 sacks) is added and continuously stirred with a bamboo inner stalk until it reaches the critical point (saturation)

where salt no longer dissolves.

50. It is made the day before and used the next day after skimming off the impurities that float to the top.

(2) Why the Critical-Point Method Is Stable

Because salinity and dissolving speed differ depending on the type of salt (refined salt, clay-pan salt, sun-dried salt, etc.), the critical-point method is stable.

51. Using only the dissolved brine achieves 18 Baumé (the appropriate salinity).
52. Sprinkling more salt on top risks over-salting, so caution is needed.
53. In the workshop, the instructor dilutes and provides brine she prepared in advance to participants.

6. Wisdom and Creativity Drawn from Nature

(1) The Connection Between Cocklebur and Velcro

54. Cocklebur is a plant whose seeds have hook-shaped protrusions that cling to the bodies of passing people or animals.
55. The story that Velcro shoes were invented based on this principle was used to emphasize that nature is a source of creative ideas.

(2) Korea's Diverse Seeds and Creativity

56. Unlike Japan, where a given region may have only one or two of the same kind of bean seed, Korea holds diverse seeds, including about 300 native legume varieties.
57. It was emphasized that this diversity and the ability to respond creatively even in difficult environments have advanced Korean food culture.

Part 2: Summary

This is a hands-on workshop covering pea-based meju production and the experience of making makjang with it. It shares detailed know-how on adapting traditional methods to modern life—how to make small meju in summer without risk of spoilage, managing temperature and humidity during fermentation, and distinguishing good mold from bad mold. In particular, it explains the principle of the meju's “self-heating phenomenon” and introduces how to use various wrapping materials such as pine needles, lotus leaves, and cocklebur instead of straw.

Participants carry out hands-on practice making makjang by mixing pea meju with aged doenjang, fermented barley flour, glutinous-rice porridge, and so on. In this process, the instructors emphasize the deep flavor and health benefits of traditional jang—particularly its value as a natural food without additives—explain the differences from commercial products, and stress the importance of taste education. Fermentation expert Jason White highlights the historical value and crisis of Korean fermentation culture, arguing that tradition must be preserved while evolving fermentation culture through scientific inquiry and creative approaches. The lecture aims to create a jang-making culture that anyone can easily follow through experimentation and record-keeping, and suggests the potential for developing new jang and sauces using various seasonal beans and local ingredients (mango, banana, etc.).

1. Pea Meju Production and Fermentation Management

(1) Making and Steaming Small Meju

- 58. Because of the spoilage risk in summer, instead of the traditional large meju, peas are steamed for 20–30 minutes to make small meju. Peas sprinkled with water are placed in a steamer and, once steam rises, steamed a little longer to become soft and moist.
- 59. Small meju, being small, can go straight into fermentation without the surface-drying stage.

(2) Ingredients and Environment for Meju Fermentation

- 60. Use of chojae: Instead of traditional straw, broad leaves such as pine needles, lotus leaves, rice straw, stonecrop leaves, cocklebur, and kudzu leaves are used. These lift the meju off the floor for air circulation (airborne fermentation) and prevent moisture evaporation and insect intrusion. They are not used to fully wrap but to line the bottom or cover.
- 61. Temperature and humidity management: Temperature and humidity are key to meju fermentation. If the meju dries out too much, fermentation cannot be properly controlled, so maintaining appropriate humidity is essential. When humidity is high due to rain, brief microwaving can remove moisture.
- 62. Self-heating phenomenon: During fermentation, the meju generates its own heat to expel moisture; this is a normal process and requires no worry.

(3) Home Meju Management (Surface-Drying and Moist Interior)

63. Step 1 (surface-drying): Unwrap the meju brought home and dry the surface crisp in sunlight and natural wind. Keep the meju spaced apart so they do not touch.
64. Step 2 (wrapping and aging): Once it reaches the “dry surface, moist interior” state, wrap it again in stonecrop leaves or wrapping paper to keep moisture from escaping. This forms a protective barrier against harmful external microbes and aids gentle fermentation.

(4) Distinguishing Mold

65. Good mold: At first, a white mold (“pear blossom”) appears like heat haze, followed by golden mold (“golden blossom”) appearing together.
66. Bad mold: Deep-yellow mold from humid, dark environments, and black or blue mold that forms in cold and damp conditions, should be avoided.

(5) Meju Size and Fermentation Speed

Making meju small makes fermentation easier and faster (about one week), allowing tasty jang to be made.

2. Makjang Production and Use

(1) Makjang Ingredients and Recipe

Base ingredient: well-fermented pea meju, ground for use.

(2) Flavor-Enhancing Ingredients

67. Seed doenjang (aged doenjang): Mixing in 10-year-aged doenjang brings out depth even with small meju. Using soy sauce instead of aged doenjang gives a similar effect.
68. Fermented barley flour: Flour made by steaming barley, aging it for three years, and grinding it; it adds a savory, refined aroma and flavor.
69. Glutinous-rice porridge: As the glutinous-rice starch breaks down, it raises the sweetness, making a very tasty jang.
70. Others: shiitake powder (umami), oilseed-residue powder, red-pepper powder, and so on are added.

(3) Salinity Adjustment

71. It is important to adjust jang salinity according to temperature and season rather than region. With nationwide warming, 18 Baumé is common, but in the hot summer it may be raised to 20 Baumé.
72. For refrigerated storage, lowering the salinity to 16% is

acceptable.

(4) Using and Aging Makjang

73. The freshly made jang is safe to eat immediately, but aging it for about a week gives a deeper flavor.
74. It can be eaten as a ssamjang for cabbage wraps, or used as a dip by blending in seasonal fruits such as mango, banana, or soft persimmon. This sauce pairs well with vegetable tempura and fish dishes.

3. The Value and Philosophy of Traditional Jang

(1) The Excellence and Health Benefits of Traditional Jang

75. Korean-style soy sauce made from naturally fermented meju leaves a strong umami and long finish behind its saltiness, and is healthful as it is made only from meju, sun-dried salt, and water without additives.
76. It reduces the thirst that comes from eating additive-heavy foods; there is also a case of a patient with stomach discomfort from radiation therapy soothing their stomach with soy-sauce tea.

(2) The Importance of Taste Education

It is important to let children experience good jang flavor from an early age to develop their ability to discern taste. We must reclaim palates accustomed to large corporations' additive-laden quick jang and protect the value of our food.

(3) A Philosophy of Fermentation (Jason White)

77. Fermentation is a process of “collaboration” combining microbes, nature, the environment, and human intention.
78. Korea's fermentation tradition is a precious heritage that was nearly lost through the colonial period and industrialization; it must be preserved and carried on in a modern way.
79. Rather than following past methods exactly, an attitude of advancing them to suit modern life through scientific inquiry and creative approaches is needed.

(4) The Importance of Recording and Experimentation (Ji Hyun-suk)

- This workshop aims to experiment and research together and to record the process. By measuring and recording things such as weight change immediately after molding and after fermentation, anyone can contribute to creating a standardized jang-making process in which success is

achievable for all.

4. Use and Application of Various Ingredients

(1) Use of Various Beans

Besides peas, distinctive jang can be made by making meju from various seasonal beans such as lentils and mung beans.

(2) Uses for Pea Pods

80. Taegakjang (pea-pod soy sauce): Filling a glass jar with brine and pea pods produces a clear soy sauce.
81. Pea-husk tea: The pods can be dried and steeped as tea, or used as stock when making jang to deepen the flavor.
82. Kombucha making: Adding sugar to the husk tea and fermenting it makes tasty kombucha.

Q&A

- In response to the concern that the antibacterial function of pine needles might inhibit fermentation, it was answered that since they are used only to allow air circulation rather than to fully wrap the meju, there is no problem.

Assignments

83. Unwrap the pea meju made in the workshop (six walnut-sized pieces) and carry out “surface-drying” in sunlight and wind; once it reaches the “dry surface, moist interior” state, wrap it again in cocklebur or lotus leaves and ferment it for one week in a warm, sunny spot.
84. Measure the weight immediately after meju production and after fermentation is complete, and record the change.
85. Using the provided ingredients (split pea meju, aged doenjang, barley flour, etc.), complete the makjang-making.
86. At home, make your own makjang using glutinous-rice porridge, aged doenjang, and so on.
87. Soak soybean husks (taegak) in brine to make husk soy sauce, and dry pea husks to drink as tea or use as stock for jang-making.
88. Compare the ingredient labels and taste of commercial soy sauce and traditional soy sauce, and reflect on their value.
89. Participate in the next workshop (barley meju-making) to continue learning.

